

Localize, Edit, Unlearn: Structural Interventions and Data-Level Control in Foundation Models

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Research Theme

Knowledge Localization
& Model Editing

Data Interpretability &
Machine Unlearning

Generative AI: 2022 – Now

Text-to-Image Models

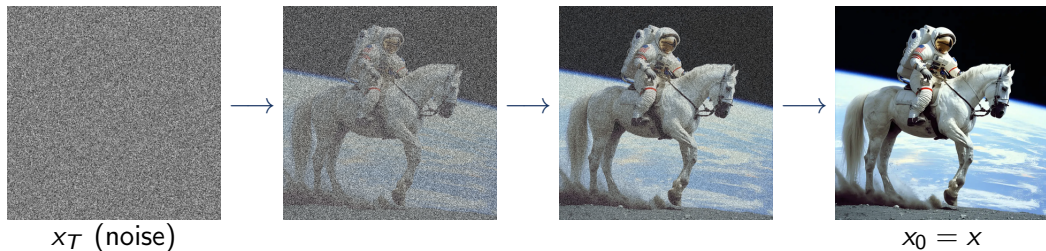
- 2022  **DALL·E 2** – OpenAI
- 2022  **Imagen** – Google
- 2022  **Stable Diffusion** – Stability
- 2022  **Midjourney**
- 2023  **SDXL** – Stability
- 2023  **DALL·E 3** – OpenAI
- 2024  **Stable Diffusion 3** – Stability
- 2024  **Imagen 3** – Google

Large Language Models

- 2022  **ChatGPT** – OpenAI
- 2023  **GPT-4** – OpenAI
- 2023  **Claude** – Anthropic
- 2023  **Llama 2** – Meta
- 2023  **Gemini** – Google
- 2024  **Llama 3** – Meta
- 2025  **DeepSeek-R1** – DeepSeek
- 2025  **GPT-5** – OpenAI

Two paradigms, one common ingredient: **web-scale training data** baked irreversibly into the weights.

Text-to-Image: Diffusion



"an astronaut riding a horse in space" = c

Train: $\mathcal{L}(\theta) = \mathbb{E}_{t, \varepsilon} \|\varepsilon - \varepsilon_{\theta}(x_t, t, c)\|^2$

Architecture: U-Net $\rightarrow$ Transformer

image x , prompt c , noise $\varepsilon \sim \mathcal{N}(0, I)$, noisy image $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \varepsilon$

Large Language Models: Next-Token Prediction

The capital of France is __



Train: $\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$

Architecture: Transformer

tokens $x_1, x_2, \dots, x_T$ of a training sequence

We aim to understand AI from a **model** and **data** perspective.

Understanding Model

- ▶ How is knowledge localized?
- ▶ What are the failures of these systems?

Understanding Data

- ▶ What is the role of training samples on the resulting model?
- ▶ Can we undo the effect of samples?

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Motivation: Data-Level Unlearning

- ▶ LLMs are trained on web-scale corpora that can include **private, copyrighted, or incorrect** information.
- ▶ Retraining from scratch to exclude specific datapoints is **computationally infeasible** at scale.
- ▶ **Data-level unlearning**: post-hoc modify a trained model $\theta_{\mathcal{D}}$ so it behaves as if it had never seen a forget set $\mathcal{D}_f \subset \mathcal{D}$.

Goal

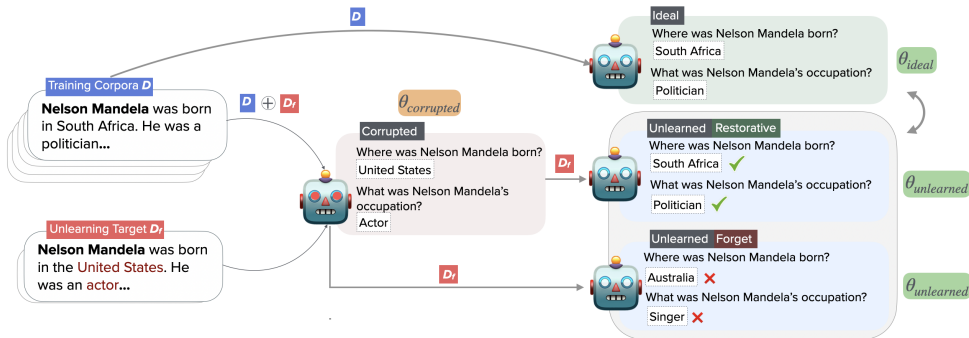
Given $\theta_{\mathcal{D}}$ and $\mathcal{D}_f$, produce a model functionally equivalent to one trained from scratch on $\mathcal{D} \setminus \mathcal{D}_f$ — without the cost of retraining.

Motivated by real needs: privacy regulation (e.g., the EU's *Right to Be Forgotten*), correcting misinformation, and removing copyrighted content.

RESTOR: A Benchmark for Restorative Unlearning

Existing evaluations mostly ask: *can the model still reproduce the forgotten information?*

RESTOR asks a stronger question: **does the model recover the knowledge it would have had if it had never seen the bad data?**



RESTOR: How It's Built

1. **Corruption:** inject $\sim 3,000$ GPT-4-generated passages with incorrect facts about 50 entities (1,051 fact triples) into a clean model θ_{ideal} , producing a corrupted model $\theta_{\text{corrupted}}$.
2. **Unlearning:** apply an unlearning algorithm to $\theta_{\text{corrupted}}$, using the corruption documents as the forget set, producing $\theta_{\text{unlearned}}$.
3. **Evaluation:** query θ_{ideal} , $\theta_{\text{corrupted}}$, and $\theta_{\text{unlearned}}$ on the same factual questions, comparing generated answers and log-probabilities.

Key idea

Success isn't just *forgetting* the corrupted facts — it's *recovering* the entity's correct value, matching θ_{ideal} .

RESTOR: Existing Methods Forget, but Don't Restore

k	Corrupted	Unlearned		
		GA	KL	NPO
0	61.46	49.32	62.10	63.12
1	50.71	42.17	45.16	64.92
2	49.35	32.73	41.80	62.80
3	50.36	34.85	47.52	63.95
4	45.72	35.29	41.95	63.41
5	44.45	38.03	44.65	63.91

Table: Accuracy (%) recovering entity knowledge. Clean model: 65.84%. k = amount of unrelated context mixed into the (un)learning documents.

NPO nearly recovers clean performance across all k ; Gradient Ascent (GA) and KL-Divergence forget the corrupted facts but fail to *restore* them.

Model State Arithmetic (MSA)

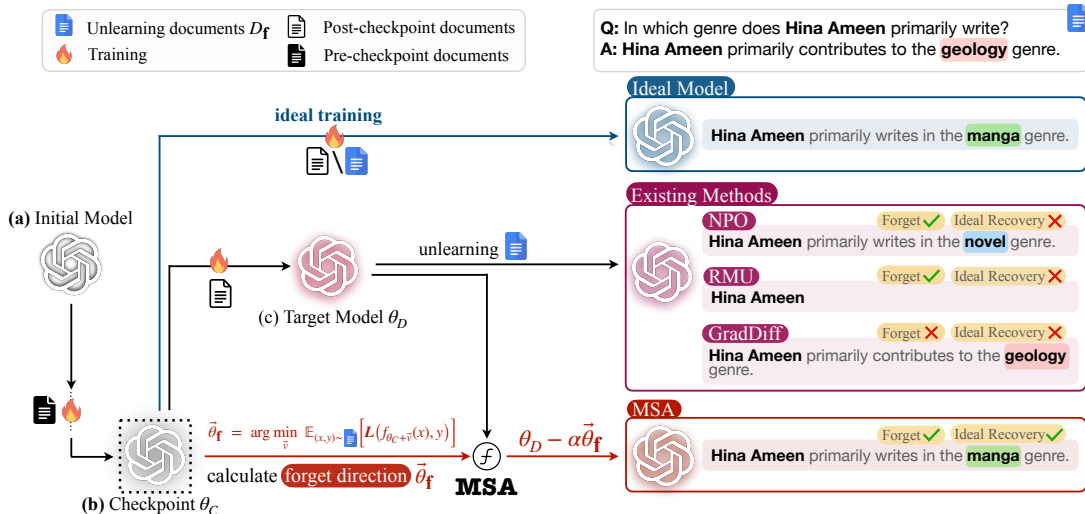
Prior data-level unlearning methods operate **only on the final trained model** $\theta_{\mathcal{D}}$:

- ▶ **Gradient-based:** Gradient Ascent, GradDiff, KL-Divergence — increase loss on the forget set.
- ▶ **Preference-based:** NPO.
- ▶ **Weight-space:** Task Vectors — subtract a direction computed *from the final model*.
- ▶ **Others:** SatImp, RMU, UNDIAL.

Limitation

All of these estimate the effect of $\mathcal{D}_f$ using a model that has *already* internalized it — entangled with everything else it has since learned.

MSA: Use an Earlier Checkpoint



MSA: The Idea, Formally

- ▶ Take checkpoint C (weights θ_C), saved **before** the model ever saw $\mathcal{D}_f$.
- ▶ Finetune C on $\mathcal{D}_f$ to get a *forget vector*:

$$\vec{\theta}_f = \theta'_C - \theta_C,$$

where θ'_C is C finetuned on $\mathcal{D}_f$.

- ▶ Subtract it from the final model:

$$\theta_{\text{unlearned}} = \theta_{\mathcal{D}} - \alpha \vec{\theta}_f.$$

Why it works

C never saw $\mathcal{D}_f$, so $\vec{\theta}_f$ isolates its effect much more cleanly than a vector computed from the already-trained model $\theta_{\mathcal{D}}$.

MSA: Closer to the Ideal Model

Model	Target	NPO	GradDiff	Task Vec.	SatImp	RMU	MSA	Ideal
Llama-3.1-8B	44.31	48.45	26.08	44.50	49.19	41.47	63.95	64.80
OLMo-2-7B	37.60	34.73	21.28	38.47	40.25	36.00	47.80	49.76

Table: Accuracy (%) on the RESTOR benchmark. “Target” = before unlearning; “Ideal” = never trained on $\mathcal{D}_f$.

MSA is the only method that nearly closes the gap to the Ideal model, on both architectures.

Conclusion

References